
GENERAL RELATIVITY TUTORIAL

Problem set 3, 25.09.2020.

■ PROBLEM 7 Surfaces and manifolds.

Consider the following surfaces imbedded in 3-dimensional Euclidean space: (i) a sphere, (ii) a torus, (iii) surface of a cone, shown in Figure 2.

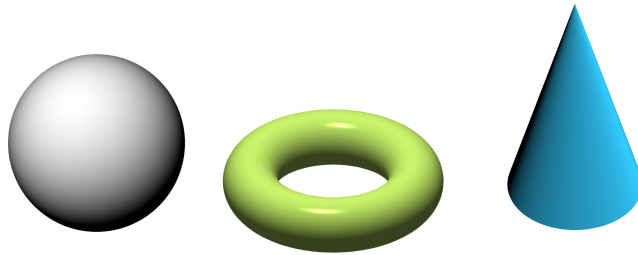


Figure 1: From left to right: a sphere, a torus, a cone.

- (a) Construct a set of charts (atlas) for each of these surfaces, i.e. construct coordinate systems on these surfaces by expressing them in terms of 3-dimensional Cartesian coordinates. What can you say about the tip of the cone (in the context of differentiable manifolds)?
- (b) Find what an infinitesimal line element is on these surfaces (in one particular chart).
Hint: Remember that for 3-dimensional Euclidean space the line element in Cartesian coordinates is just $dl^2 = dx^2 + dy^2 + dz^2$.

SOLUTION (a) Sphere. *Sterographic projection* of a sphere is an example of chart. Imagine the sphere of diameter D and an infinite plane, such that the sphere touches the plane at one point, which one can call the south pole S . Draw a straight line starting at the north pole N . This line intersects the sphere once and the plane also once (assuming we do not count the north pole). This point is at the radial distance r from S given by, $r = D \sin(\theta)$, where $0 \leq \theta < \pi/2$ is the angle with the vertical line connecting the north and south poles. The position on the plane is then given by $(x, y) = (D \sin(\theta) \cos(\varphi), D \sin(\theta) \sin(\varphi))$, where $0 \leq \varphi < 2\pi$ is the angle between the x axis and the line. The map $S^2(D/2) \rightarrow R^2$ is a smooth (C^∞) chart that covers the sphere minus the north pole. Because the point N is not covered by the chart, we need another chart. One such chart is stereographic projection of the sphere onto the plane which touches it at the north pole. This chart covers the sphere minus the south pole. Clearly, in the overlapping regions (the sphere minus both poles) the maps overlap smoothly (there is a smooth bijective map between the two planes which maps each point on one plane once and precisely once onto each point on the other plane. One example of such a smooth map is a set of vertical lines, which punctures each plane exactly once. (Can you think of another map?)

Torus. A torus with radii R and $r < R$ can be coordinated as follows,

$$\begin{aligned}x &= [R + r \cos(\theta)] \sin(\varphi) \\y &= [R + r \cos(\theta)] \cos(\varphi) \\z &= r \sin(\theta) .\end{aligned}\tag{7.1}$$

An example of the coordinate ranges that fully covers the torus is, $\theta \in (0, 2\pi]$, $\varphi \in (0, 2\pi]$, which means that this covering is not a chart, since it is not open at $\theta = 2\pi$ with φ arbitrary, and also at $\varphi = 2\pi$ with θ arbitrary. Now, it is not hard to see that one needs at least four charts to cover the torus. One needs two charts in the φ direction (*e.g.* one with $\varphi \in (0, 2\pi)$, another with $\varphi \in (\pi, 3\pi)$) and two charts in the θ directions. Since φ and θ are independent coordinates, that means that at least four charts are needed to cover the torus.

Cone. One example of the coordinates that cover the whole cone is the projection of the cone onto a plane which touches the cone at its tip. The set of vertical lines – each traverses the cone and the plane exactly once – is an example of a chart that completely covers it. The Cartesian coordinates are in this case given by, $x = r \cos(\alpha)$, $y = r \sin(\alpha)$, where $r = \ell \sin(\theta)$, α is the angle between the x axis and the line, ℓ is the distance from the point on the cone and its tip and θ is the cone's angle. This chart is smooth (C^∞) everywhere except at its tip, where the map is continuous. This means that the chart is continuous (C^0). (Can you rigorously prove that?)

As an exercise, sketch all the charts discussed above.

■ PROBLEM 8 Coordinate transformations.

- (a) How would the product of a covariant and a contravariant vector, $A^\mu B_\mu$, transform under general coordinate transformations?
- (b) What about the quantity $g_{\mu\nu} g^{\nu\rho} = \delta_\mu{}^\rho$?
- (c) Can you show that the derivative of a covariant vector, $\partial_\mu A_\nu$, does not transform as a tensor? Show that for linear coordinate transformations (*i.e.* Lorentz transformations) $\partial_\mu A_\nu$ is indeed a tensor.
- (d) Check that the transformation rules (4.10) and (4.13) in the lecture notes (https://webpace.science.uu.nl/~hooft101/lectures/genrel_2013.pdf) form a group, *i.e.* the transformation $x \rightarrow u$ yields the same tensor as the sequence $x \rightarrow v \rightarrow u$. Make use of the fact that partial differentiation obeys

$$\frac{\partial x^\mu}{\partial u^\nu} = \frac{\partial x^\mu}{\partial v^\alpha} \frac{\partial v^\alpha}{\partial u^\nu} .\tag{8.1}$$

SOLUTION: (a) The vector components A^μ transform as, $A^\mu \rightarrow A'^\mu = \frac{\partial x'^\mu}{\partial x^\alpha} A^\alpha$, while the components of the form transform as, $B_\mu \rightarrow B'_\mu = \frac{\partial x^\alpha}{\partial x'^\mu} B_\alpha$. Putting everything together gives,

$$A^\mu B_\mu \rightarrow A'^\mu B'_\mu = \frac{\partial x'^\mu}{\partial x^\alpha} A^\alpha \frac{\partial x^\beta}{\partial x'^\mu} B_\beta = A^\alpha B_\alpha ,\tag{8.2}$$

where we made use of, $\frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\mu} = \delta_\alpha^\beta$. This means that $A^\mu B_\mu$ transforms as a scalar under (global or local) Lorentz transformations, *i.e.* it does not transform.

(b) The corresponding (symmetric metric) tensors transform as,

$$\begin{aligned} g_{\mu\nu} &\rightarrow g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} \\ g^{\nu\rho} &\rightarrow g'^{\nu\rho} = \frac{\partial x'^\nu}{\partial x^\gamma} \frac{\partial x'^\rho}{\partial x^\eta} g^{\gamma\eta}, \end{aligned} \quad (8.3)$$

such that the product transforms as,

$$g_{\mu\nu} g^{\nu\rho} \rightarrow g'_{\mu\nu} g'^{\nu\rho} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\eta} g_{\alpha\beta} g^{\beta\eta} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\rho}{\partial x^\eta} \delta_\alpha^\eta, \quad (8.4)$$

thus it transforms as a (1,1) tensor. Since the Kronecker delta is a very special tensor, $\delta_\alpha^\eta = \text{diag}(1, 1, 1, 1)$, hence in (8.4) we can write $\frac{\partial x'^\rho}{\partial x^\eta} \delta_\alpha^\eta = \frac{\partial x'^\rho}{\partial x^\alpha}$. Inserting this into (8.4) we conclude that,

$$g_{\mu\nu} g^{\nu\rho} = \delta_\mu^\rho \rightarrow g'_{\mu\nu} g'^{\nu\rho} = \delta_\mu^\rho \quad (8.5)$$

such that $g_{\mu\nu} g^{\nu\rho}$ transforms as the unity tensor, which under (local or global) Lorentz transformations transforms into itself.

(c) $\partial_\mu A_\nu$ transforms as,

$$\partial_\mu A_\nu \rightarrow \partial'_\mu A'_\nu = \frac{\partial x^\rho}{\partial x'^\mu} \partial_\rho \left(\frac{\partial x^\sigma}{\partial x'^\nu} A_\sigma \right) = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} (\partial_\rho A_\sigma) + \frac{\partial x^\rho}{\partial x'^\mu} \left[\partial_\rho \left(\frac{\partial x^\sigma}{\partial x'^\nu} \right) \right] A_\sigma. \quad (8.6)$$

While the first term after the second equality is the transformation of a (0,2) tensor, the second (last) term is non-tensorial. For (global) Lorentz transformations, $\Lambda_\nu^\sigma = \frac{\partial x^\sigma}{\partial x'^\nu}$ is independent of coordinates, such that the second term drops out, and indeed $\partial_\mu A_\nu$ transforms as a (0,2)-tensor.

(d) The corresponding transformations are,

$$\tilde{A}_\nu(u) = \frac{\partial x^\mu}{\partial u^\nu} A_\mu(x(u)), \quad \tilde{F}_\nu(u) = \frac{\partial u^\mu}{\partial x^\alpha} F^\alpha(x(u)). \quad (8.7)$$

For the first equality, we have

$$\bar{A}_\nu(v) = \frac{\partial x^\mu}{\partial v^\nu} A_\mu(x(v)), \quad \tilde{A}_\rho(u) = \frac{\partial v^\nu}{\partial u^\rho} \bar{A}_\nu(v(u)). \quad (8.8)$$

Upon composing the two one gets,

$$\tilde{A}_\rho(u) = \frac{\partial v^\nu}{\partial u^\rho} \frac{\partial x^\mu}{\partial v^\nu} A_\mu(x(v(u))) = \frac{\partial x^\mu}{\partial u^\rho} A_\mu(x(u)), \quad (8.9)$$

where in the last step we made use of the chain rule (8.1).

Analogously, for the second transformation in (8.7) we have,

$$\bar{F}^\mu(v) = \frac{\partial v^\mu}{\partial x^\alpha} F^\alpha(x(v)), \quad \tilde{F}^\sigma(u) = \frac{\partial u^\sigma}{\partial v^\mu} \bar{F}^\mu(v(u)). \quad (8.10)$$

Upon composing the two transformations we get,

$$\tilde{F}^\sigma(u) = \frac{\partial u^\sigma}{\partial v^\mu} \frac{\partial v^\mu}{\partial x^\alpha} F^\alpha(x(v(u))) = \frac{\partial u^\sigma}{\partial x^\alpha} F^\alpha(x(u)). \quad (8.11)$$

What we have learned from the above exercise is that, firstly general coordinate transformations (local Lorentz transformations) of vectors and forms obey the composition rule, *i.e.* if g_1 and g_2 are elements of the group, then $g = g_2 \cdot g_1$ is also an element of the group (the closure property). Next we have that the Kronecker delta tensor δ_μ^ρ is the unity element of the group, *i.e.* $g \cdot \delta = g = \delta \cdot g$ for g an arbitrary coordinate transformation. Finally, to show that for each g , there exists g^{-1} such that, $g \cdot g^{-1} = \delta = g^{-1} \cdot g$, recall that the inverse of $\frac{\partial x^\mu}{\partial u^\rho}$ is $\frac{\partial u^\rho}{\partial x^\mu}$. Indeed, by the chain rule we have, $\frac{\partial x^\mu}{\partial u^\rho} \frac{\partial u^\rho}{\partial x^\nu} = \frac{\partial x^\mu}{\partial x^\nu} = \delta_\nu^\mu$ and similarly, $\frac{\partial u^\rho}{\partial x^\mu} \frac{\partial x^\mu}{\partial u^\sigma} = \frac{\partial u^\rho}{\partial u^\sigma} = \delta_\sigma^\rho$. Taking everything together, we conclude that general coordinate transformations form a group.

■ PROBLEM 9 Symmetrizing and antisymmetrizing tensors.

For any 2-tensor $T_{\mu\nu}$ (in four dimensions) we define the tensors

$$T_{(\mu\nu)} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu}) , \quad T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}) . \quad (9.1)$$

These are, respectively, called symmetric and antisymmetric part of the tensor $T_{\mu\nu}$.

(a) Is it true that for any tensor $A_{\mu\nu} = A_{(\mu\nu)} + A_{[\mu\nu]}$? How many components do $A_{(\mu\nu)}$ and $A_{[\mu\nu]}$ have?

(b) If $B_{\mu\nu} = B_{\nu\mu}$ and $C_{\mu\nu} = -C_{\nu\mu}$, prove that (A is a generic tensor)

$$A_{\mu\nu}B^{\mu\nu} = A_{(\mu\nu)}B^{\mu\nu} , \quad A_{\mu\nu}C^{\mu\nu} = A_{[\mu\nu]}C^{\mu\nu} . \quad (9.2)$$

For higher rank tensors, this concept can be generalized to obtain totally symmetric and anti-symmetric tensors by definitions

$$T_{(\mu_1\mu_2\ldots\mu_k)} = \frac{1}{k!} \sum_{\sigma \in S^k} T_{\sigma(\mu_1)\sigma(\mu_2)\ldots\sigma(\mu_k)} , \quad (9.3)$$

$$T_{[\mu_1\mu_2\ldots\mu_k]} = \frac{1}{k!} \sum_{\sigma \in S^k} (-1)^{\pi(\sigma)} T_{\sigma(\mu_1)\sigma(\mu_2)\ldots\sigma(\mu_k)} , \quad (9.4)$$

respectively, where T is an arbitrary covariant tensor of rank k . In (9.3) and (9.4), S^k denotes the group of k objects, and π the so-called parity of the permutation, with $\pi(\sigma) = 0$ if σ is an even permutation, and $\pi(\sigma) = 1$ if it is an odd permutation.

(c) Consider the case of an arbitrary 3-tensor $T_{\mu\nu\rho}$. How many components does it have? Write explicitly the form of $T_{(\mu\nu\rho)}$ and $T_{[\mu\nu\rho]}$.

(d) For an arbitrary 3-tensor $T_{\mu\nu\rho}$ is it true that $T_{\mu\nu\rho} = T_{(\mu\nu\rho)} + T_{[\mu\nu\rho]}$? Prove your result.

SOLUTION: (a) It is true. In n dimensions, $T_{(\mu\nu)}$ is $n(n+1)/2$ components, $T_{[\mu\nu]}$ has $n(n-1)/2$ components, such that together they have n^2 components. The difference in the number of components can be explained by noting that $T_{(\mu\nu)}$ has n independent components on the diagonal, while the diagonal components of $T_{[\mu\nu]}$ are all zero. This means that $T_{(\mu\nu)}$ has n components more than $T_{[\mu\nu]}$, which can be used to derive the above quoted result.

(b) The proofs can be constructed as follows,

$$\begin{aligned} A_{\mu\nu}B^{\mu\nu} &= [A_{(\mu\nu)} + A_{[\mu\nu]}] B^{(\mu\nu)} = A_{(\mu\nu)}B^{(\mu\nu)} + A_{[\mu\nu]}B^{(\mu\nu)} , \\ A_{\mu\nu}C^{\mu\nu} &= [A_{(\mu\nu)} + A_{[\mu\nu]}] C^{[\mu\nu]} = A_{[\mu\nu]}C^{[\mu\nu]} + A_{(\mu\nu)}C^{[\mu\nu]} . \end{aligned} \quad (9.4)$$

The second equalities follow from the observation that the product of a symmetric and anti-symmetric tensor must vanish. The sum must be both symmetric and antisymmetric under the exchange of two indices which means that it must vanish. To see this explicitly, observe that for two general tensors, $(B^{\mu\nu} - B^{\nu\mu})(A^{\mu\nu} + A^{\nu\mu}) = B^{\mu\nu}A^{\mu\nu} - B^{\nu\mu}A^{\mu\nu} + B^{\mu\nu}A^{\nu\mu} - B^{\nu\mu}A^{\nu\mu} = 0$. Now, the first and the last term are equal in magnitude but opposite in sign. Analogously, the second and the third term are equal in magnitude, opposite in sign, such that the total result is zero.

(c) and **(d)** An arbitrary symmetric (anti-symmetric) 3-tensor has $n(n+1)(n+2)/6$ ($n(n-1)(n-2)/6$) components. These tensors can be explicitly written as,

$$\begin{aligned} T_{(\mu\nu\rho)} &= \frac{1}{6} [T_{\mu\nu\rho} + T_{\nu\rho\mu} + T_{\rho\mu\nu} + T_{\rho\nu\mu} + T_{\mu\rho\nu} + T_{\nu\mu\rho}] \\ T_{[\mu\nu\rho]} &= \frac{1}{6} [T_{\mu\nu\rho} + T_{\nu\rho\mu} + T_{\rho\mu\nu} - T_{\rho\nu\mu} - T_{\mu\rho\nu} - T_{\nu\mu\rho}] . \end{aligned} \quad (9.4)$$

This also means that the total number of components in these two tensors does not sum up to n^3 , which means that such a decomposition is in general incomplete (it is complete only for 2-tensors). For example, $T_{(\mu\nu)\rho}$ is neither symmetric nor antisymmetric with respect to the pairs of indices $\{\mu, \rho\}$ or $\{\nu, \rho\}$. It turns out that adding a tensor that is symmetrized with respect to two indices and antisymmetrized with respect to another two indices completes the decomposition of 3-tensors. A more general technique for decomposition of higher rank tensors is based on Young diagrams and Young tableaux.